

Creating a Theoretical Map of the Distribution of Vicuña

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***Vicugna vicugna* habitat preferences are:**

Elevations: 3500 - 5700 masl (meters above sea level)

Mean precipitation: less than 2000 mm.

Mean temperature: less than 37°C

Low vegetation productivity: NDVI values between 200 and 500

I have following variables (layers in the data folder):

Elevation: ELEVATION.rst (in masl)

Productivity (NDVI): NDVI_ANNUAL_MEAN.rst

Temperature: TEMPERATURE_MEAN.rst (in °C)

Precipitation: ANNUAL_PRECIPITATION.rst (in mm)

Questions

3. In the Geographic range description from the Animal Diversity Web it says that the range is distributed in Southern Peru, western Bolivia, northwestern Argentina, and northern Chile. In our map, it is distributed along other country/countries also. Which countries are these?

- In our map, it is also distributed along Ecuador, Colombia, and Venezuela.

4. How does your distribution compare to the one described in Wikipedia?

- My distribution map looks very similar to the one described in Wikipedia. Their map had more buffers around the described area compared to mine. They also left out a small part of the northern area as seen in figure 3.

5a. How many suitable pixels are there?

- There are 16,434 suitable pixels.

5b. What % of South America is suitable for Vicuna Vicuna (as per these mentioned criteria)?

- About 2.17% of South America is suitable for Vicuna Vicuna.

6. (Optional) Complete the following table by calculating suitable habitat areas within different countries. Then, answer the questions.

Country name	No. of Suitable pixels	Suitable area (in hectare)	No. of total pixels (both suitable and unsuitable)	% suitable area for the habitat
Peru	9,453	26,019,861	53,338	17.72
Bolivia	4,807	13,231,511	46,272	10.39
Chile	136	374,347	37,610	0.36
Argentina	1,132	3,115,887	139,263	0.81

6a. Which country has the largest suitable area for Vicuna Vicuna?

- Peru has the largest suitable area of Vicuna Vicuna.

6b. What is the total suitable area in square kilometers (overall in all countries)?

- The total suitable area of all countries is about 45,2354 square kilometers.

6c. What function is performed by Zonal Statistics as Table tool (i.e. local, focal, zonal or global)? Why?

- The Zonal Statistics as Table tool can calculate statistics like mean, median, etc for the values within zones defined by another dataset. These statistics are outputted in a table format. The summarizing data within defined zones allows for comparisons between different areas. They are also a better way to organize and analyze the results, rather than a raster output.

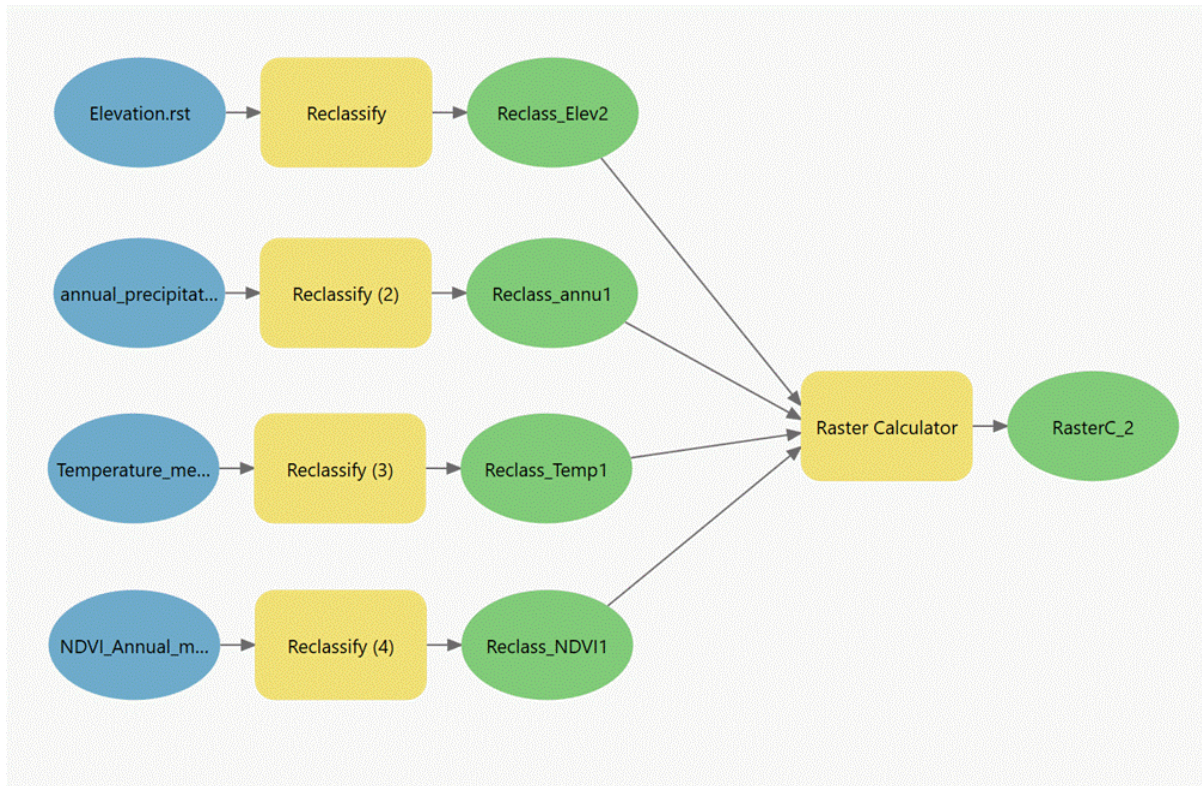


Figure 1: A complete cartographic model diagram before running.

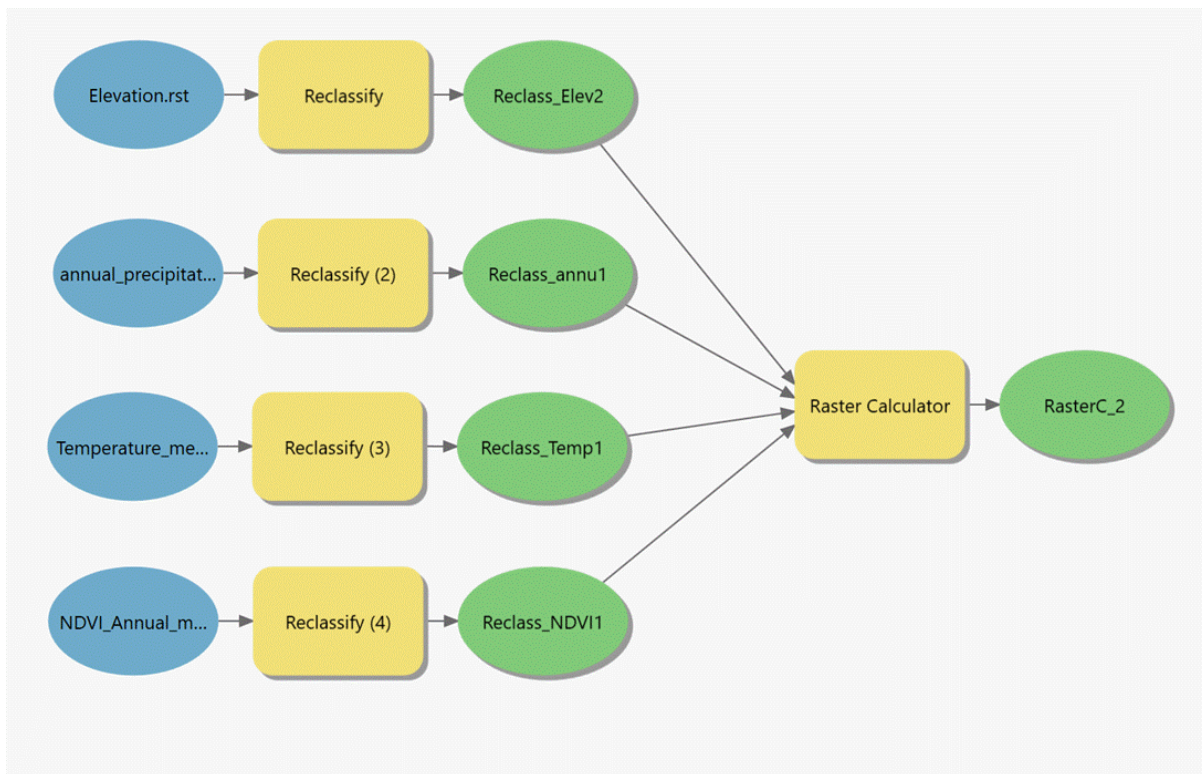


Figure 2: A complete cartographic model diagram after running.



Figure 3: Distribution of Vicuña, left is a map from Wikipedia and on the right is mine.